PROBLEM SET 8 DUE 5.24

1. Let k be a field and let $X \subseteq \mathbb{A}_k^n$. Show that $Z(I(X))$ is the Zariski-closure of X in $\mathbb{A}_k^n$.

2. Let k be a field and $n \geqslant 1$. Show that $\mathbb{A}_k^n$ is compact in the Zariski topology.

3. Let k be an infinite field and $n \geqslant 1$. Show that $I(\mathbb{A}_k^n) = 0$.

4. Let k be an infinite field and $n \geqslant 1$. Show that $\mathbb{A}_k^n$ is *not* the union of two proper Zariski-closed subsets.

5. Let k be a field and $n \geqslant 1$. Show that the following statements are equivalent:

(1) If $\mathfrak{a} \subsetneq k[x_1, \ldots, x_n]$ is a proper ideal, then $Z(\mathfrak{a}) \neq \varnothing$.
(2) If $\mathfrak{m} \subsetneq k[x_1, \ldots, x_n]$ is a maximal ideal, then $Z(\mathfrak{m}) \neq \varnothing$.
(3) If $Z(\mathfrak{a}) = \varnothing$ for some ideal $\mathfrak{a} \subseteq k[x_1, \ldots, x_n]$, then $\mathfrak{a} = (1)$.
(4) Every maximal ideal of $k[x_1, \ldots, x_n]$ has the form $(x_1 - a_1, \ldots, x_n - a_n)$ for some $(a_1, \ldots, a_n) \in \mathbb{A}_k^n$.

In particular, all of the above statements hold for an algebraically closed field.

6. Let A be a ring and S be a multiplicative subset of A contained in $A^\times$. Show that the localization map $\varphi : A \to S^{-1}A$; $a \mapsto \frac{a}{1}$ is an isomorphism.

7. Let A be a ring, $f \in A$. Show that $A_f \cong A[x]/(xf - 1)$.

8. Let $f : A \to B$ be a ring map and let S, T be multiplicative subsets of A, B respectively such that $f(S) \subseteq T$. Show that there exists a ring map

$$S^{-1}A \longrightarrow T^{-1}B$$

which sends $\frac{a}{s}$ to $\frac{f(a)}{f(s)}$.

9. Let A be a ring, $S \subseteq A$ be a multiplicative subset, and $\varphi : A \to S^{-1}A$ be the localization map. Let $\mathfrak{a}$ be an ideal of A such that $\mathfrak{a} \subseteq \ker \varphi$. Put

$$\bar{A} = A/\mathfrak{a} \quad \text{and} \quad \bar{S} = \{\bar{s} : s \in S\} \subseteq \bar{A}.$$

Prove the following statements:

(1) There exists a ring isomorphism $S^{-1}A \to \bar{S}^{-1}\bar{A}$ which sends $\frac{a}{s}$ to $\frac{\bar{a}}{\bar{s}}$.
(2) If $\bar{S} \subseteq \bar{A}^\times$, then $S^{-1}A \cong \bar{A}$.
(3) Let $n \geqslant 2$ be an integer and p a prime. Then

$$(\mathbb{Z}/n\mathbb{Z})_{\bar{p}} \cong \mathbb{Z}/m\mathbb{Z}$$

where $n = p^e m$ for some $e \geqslant 0$ and $p \nmid m$.

10. Recall the local ring R of germs of continuous functions at 0 on the real line:

$$R = \left\{ [(U, f)] : U \text{ open nbhd of } 0 \text{ in } \mathbb{R},\ f : U \to \mathbb{R} \text{ continuous} \right\}$$

where $[(U, f)] = [(V, g)]$ if and only if there exists an open neighborhood W of 0 contained in $U \cap V$ such that $f|_W = g|_W$.

Let $A = \mathscr{C}(\mathbb{R})$ be the ring of real-valued continuous functions on $\mathbb{R}$, and define

$$\mathfrak{a} \coloneqq \left\{ f \in \mathscr{C}(\mathbb{R}) : \exists \epsilon > 0 \text{ s.t. } f|_{[-\epsilon,\epsilon]} = 0 \right\}$$
$$\mathfrak{m} \coloneqq \left\{ f \in \mathscr{C}(\mathbb{R}) : f(0) = 0 \right\}$$

It is not difficult to see that $\mathfrak{a} \subseteq \mathfrak{m}$ are ideals of A and $\mathfrak{m}$ is maximal.

Show that

$$A/\mathfrak{a} \cong A_{\mathfrak{m}} \cong R.$$

11. Let A be a ring and $S, T \subseteq A$ be multiplicative subsets.

Show that the following rings are isomorphic

$$(ST)^{-1}A \cong V^{-1}(S^{-1}A) \cong U^{-1}(T^{-1}A).$$

where ST is the multiplicative subset of A generated by S and T, V is the image of T in $S^{-1}A$, and U is the image of S in $T^{-1}A$.